

University of Toronto Mississauga

Mathematical and Computational Sciences

MAT102H5S - Make-up Test

Test Booklet

Duration - 110 minutes

No Aids Permitted

Surname/Family Name: _____
First/Given Name: _____
Student Number: _____
UTORid: _____
Email: _____

This exam contains 6 pages (including this cover page) and 5 problems. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 5 (page 6). No aids are permitted on this examination. Examples of illegal aids include, but are not limited to textbooks, notes, calculators, or any electronic device.

Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 6). When filling out the scantron, ensure that bubbles are completely filled and make complete erasures. Any failure to follow these instructions which results in the scanner not being able to read your work will result in a 3 mark deduction. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 4 must be written in the space provided. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this test.**

Unless otherwise indicated, you are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work** in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work, will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- You may use any result mentioned explicitly in the readings, assignments, or in-class worksheets without proof. To use a result which is not covered by the aforementioned sources, you must provide a proof of the result as part of your work.

The Multiple Choice Booklet is printed one-sided, and may thus be used for rough work.

1. Define a relation on subsets of $\mathbb{R}$, wherein we say that $A \preceq B$ if for every $x \in A$ there is a $y \in B$ such that $x \leq y$.

(i) (1 point) Let $A = \mathbb{Z}$. Find a set $B \subseteq \mathbb{R}$ such that $A \preceq B$ and justify your answer, or argue that no such set can exist.

(ii) (3 points) Determine whether $\preceq$ is transitive.

(iii) (3 points) Determine whether $\preceq$ is left-total.

2. (5 points) Define a function $f : \mathbb{Z}_{2025} \rightarrow \mathbb{Z}_{2025}$ by $f([x]_{2025}) = [2x]_{2025}$. You may assume, without proof, that this function is well-defined. Determine whether f is injective, surjective, both, or neither.

3. (5 points) Show that for all odd positive integers n , $8 \mid (n^2 - 1)$.

4. For each $n \in \mathbb{Z}^*$, define the set $\mathcal{A}_n$ to be the set of all subsets of $\mathbb{Z}^+$ whose elements sum up to n ; that is,

$$\mathcal{A}_n = \left\{ A \subseteq \mathbb{Z}^+ : \sum_{k \in A} k = n \right\}.$$

For example, $\mathcal{A}_6 = \{\{6\}, \{5, 1\}, \{4, 2\}, \{1, 2, 3\}\}$. Note that we take the empty sum to be zero.

- (i) (3 points) Argue that $\mathcal{A}_n$ is finite. Hint: Show that $\mathcal{A}_n \subseteq \mathcal{P}(\{1, 2, \dots, n\})$.

- (ii) (3 points) If $F = \{A \subseteq \mathbb{Z}^+ : A \text{ is finite}\}$, show that $F = \bigcup_{n=0}^{\infty} \mathcal{A}_n$.

- (iii) (1 point) Conclude that F is countable.